Opinion Pooling IV: Promoting Group Epistemic Rationality

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Accuracy and Justification in Pooling

Recap: What We've Covered So Far

We've examined two families of pooling strategies:

- **Linear pooling**: Weighted averages (satisfies most axioms, but violates External Bayesianity)
- **Geometric pooling**: Geometric means over partitions (satisfies External Bayesianity, but partition-sensitive and violates lots of other axioms)

Today we're looking at how pooling relates to whether groups are **reasonable**. So we're taking more of an outsider's perspective.

Two Key Questions

How well do pooling strategies promote:

1. **Accuracy**: Getting closer to the truth
2. **Justification**: Preserving epistemic warrant

These are hopefully related; though just what the relation is isn't always clear.

Measuring Accuracy

The Accuracy-First Approach

Core idea: A credence can't be true or false, but it can be more or less *accurate*.

- If I'm 60% confident it will rain and you're 5% confident.
- If it rains: my credence was more accurate.
- If it doesn't rain: your credence was more accurate.

Can we measure accuracy of *entire credence functions*, not just single credences?

For any world w, the ideal credence function assigns:

- Credence **1** to all propositions true at w
- Credence **0** to all propositions false at w

This represents perfect knowledge of w. It's literally the God's eye view.

Accuracy = how close your credences are to the ideal

Inaccuracy = how far your credences are from the ideal

But this just raises a question, how do we measure accuracy?

Usual starting point: Sum up the "mistakes" across all your credences.

Example: You're 40% confident it will rain.

- If it rains (ideal: 100% confident), you're off by 60 percentage points
- If it doesn't rain (ideal: 0% confident), you're off by 40 percentage points

Different ways to aggregate these errors give different accuracy measures.

Annoying Technical Detail

There are some very widely accepted constraints on how you do this measurement, and a bunch of measures that satisfy those constraints.

Annoyingly, the measurement that says:

- For every proposition, you get x inaccuracy points when x is the difference between your inaccuracy and the ideal;
- You add up all the inaccuracy points across all propositions, and the lower the better;

is **not** one of them.

If you say that whenever you are off by x, you get penalised x^2 accuracy points, you do get a measure that works, and indeed the most widely used measure there is. (It's called the Breir score, after a meteorologist who proposed it in the 1950s.)

A Key Result

If you use these appropriate accuracy measures (technical details omitted, but the key moves here are largely due to Michigan faculty):

- **Probabilism vindicated**: If your credences violate probability rules, there's always a better credence function that's guaranteed to be more accurate.
- **Bayesianism vindicated**: Updating by Bayes' Rule minimizes expected inaccuracy.

This provides justification for fundamental epistemic norms.

Let's say that we measure inaccuracy by the square of the distance from perfection. Then there are two notions that get used a lot.

- **Guaranteed Inaccuracy**. Sometimes one credence function is guaranteed to be more accurate than another.
- **Expected Inaccuracy**. For any two probability functions, Pr_1 and Pr_2, we can work out how inaccurate Pr_1 expects Pr_2 to be.

Guaranteed Inaccuracy

We're interested in whether or not it will rain tomorrow.

- I have the incoherent view that it is 70% likely to rain, and 70% likely to not rain.
- Mitch has the coherent view that it is 50% likely to rain, and 50% likely to not rain.

If it rains, my inaccuracy score is 0.58. I'm off on rain by 0.3, and I'm off on not rain by 0.7, and $0.3^2 + 0.7^2 = 0.58$.

If it rains, Mitch's inaccuracy score is 0.5. He's off on rain by 0.5, and he's off on not rain by 0.5, and $0.5^2 + 0.5^2 = 0.58$.

He does better; and he also does better (for the same reason) if no rain, so he's **guaranteed** to do better.

Expected Inaccuracy

Set aside these incoherent views like having credence 0.7 in both rain and not rain. Imagine that Pr_1 and Pr_2 are coherent. They have credences x_1 and x_2 in rain, and $1 - x_1$ and $1 - x_2$ in not rain.

How accurate does each *expect* the other to be?

Since they are coherent, we'll ignore their credence in not rain, and just ask how accurate they expect the rain credence to be. (The other credence is, after all, forced.)

Expected Inaccuracy

We just pull out the standard formula for expected value.

- My expectation of your inaccuracy is given by:
- My credence in rain, times how inaccurate you would be if it rains plus
- My credence in not rain, times how inaccurate you would be if it doesn't rain.
- And your expectation of my inaccuracy is calculated the same way.

Expected Inaccuracy

Neat result:

- If we're both coherent, I will expect that I'm more accurate than you, and you will expect you're more accurate than me.
- In general, every person with a coherent set of credences will think they are maximally accurate *in expectation*.
- Of course, the person who simply guesses right about everything will be more accurate, but for any such guesser, each of us should expect to do better than them.

Accuracy-Based Arguments for Pooling

The Shakespeare Scholar Scenario

Let's start with an example. A huge community of Shakespeare scholars has credences about who authored the works attributed to Shakespeare.

- Christopher Marlowe?
- Francis Bacon?
- Edward de Vere?
- Shakespeare himself? (It was Shakespeare himself, but let's do the hypothetical.)

We want to summarize their collective view using a pooling strategy.

Suppose we pool to get group credence Q.

Problem: What if there's an alternative Q^* such that *every individual scholar* thinks Q^* would be more accurate than Q (in expectation)?

Then we've ascribed the wrong credences to the group, since there's a particular credence everyone agrees is better.

Accuracy Consensus: There should be no alternative group credence that all individuals prefer (in terms of expected accuracy).

This is like a Pareto optimality condition for credence functions.

The result: Linear pooling is the *only* strategy that satisfies Accuracy Consensus.

More precisely: if Q is not a linear pool, there exists an alternative that everyone prefers. If Q is a linear pool, no such alternative exists.

When Does This Matter?

Accuracy Consensus is important when individuals must “get behind” the group credence, e.g., in

- Representing group opinion to outsiders (the Shakespeare authorship debate, or the climate change example from last lecture); or
- Assessing group liability for collective actions (corporate responsibility cases).

If accuracy is our goal for group credences, this favors linear pooling.

Note that this doesn't require that everyone's views be **equally weighted**.

If we use linear pooling, but give one person's views 100 times as much weight, we will still not violate this constraint.

But we have to use linear pooling.

The Diversity Prediction Theorem

Estimating Numerical Quantities

Pooling strategies can aggregate any numerical estimates, not just credences.

County fair example: Lila (100 kg), Mona (94 kg), Nomy (93 kg) estimate the prize marrow's weight.

True weight: 91 kg.

In this case the **average inaccuracy** of individual estimates exceeds the **inaccuracy of the straight linear pool**:

$$\frac{1}{3}(100-91)^2+\frac{1}{3}(94-91)^2+\frac{1}{3}(93-91)^2 > \left(\frac{100+94+93}{3}-91\right)^2$$

This holds in general. The inaccuracy of the average of some estimates is lower than the average inaccuracy.

Two Options

1. Take the linear average of the estimates.
2. Pick an estimator at random, and use their estimate.

As long as the estimators differ, the inaccuracy of option 1 is **always** lower than the expected inaccuracy of option 2.

Of course, you might get lucky and pick the estimator who was closest to reality. But on average you'll do worse.

This holds *regardless* of:

- What quantities are being estimated;
- What the true values are; and
- What estimates people give

Among other things, it holds about estimates of **truth value**, and credences just are estimates of truth value.

Summary

So those are two really good features of linear pooling.

1. Linear pooling, *and only linear pooling*, always produces an outcome where there's no alternative everyone prefers.
2. Linear pooling, *and only linear pooling*, is always guaranteed to do better than just deferring to one person at random.

Deference Compatibility

When deferring to experts, we want it to be *possible* that:

1. We think experts might disagree: $P(E_1 = E_2) < 1$
2. We defer to each expert individually: $P(X \mid E_1 = p) = p$ and $P(X \mid E_2 = q) = q$
3. We defer to the pool of experts: $P(X \mid E_1 = p, E_2 = q) = F(p, q)$

Bad news: Linear pooling does *not* satisfy Deference Compatibility.

In fact, Snow Zhang recently proved a much stronger result than this.

Say that a pooling method satisfies **Strict Boundedness** if whenever the experts disagree, the pooled value is strictly between the values they propose.

Zhang proved that if there are only finite many values that the experts might announce, e.g., if they are only going to announce to four decimal places, then no pooling method satisfying **Strict Boundedness** also satisfies **Deference Compatibility**.

Discussion: The Climate Policy Advisor

Maya works for a city government with no climate science background. She consults three climate scientists about flood risk from a proposed development:

- Dr. Chen: 70% chance of severe flooding in next 20 years
- Dr. Johnson: 45% chance
- Dr. Okafor: 60% chance

Questions for discussion:

1. Should Maya adopt the linear pool (58.3%) or choose one expert to defer to entirely?
2. What if she later learns Dr. Chen has relevant local expertise? How should she update?

Bigger Picture Philosophy

Contractualism

In political philosophy, contractualism is the view that what grounds political authority is an implicit contract between the people.

It's legitimate for the state to have power because we all agree, or at least in the right circumstances we would all agree, that it is better that they have the power.

At the very least, and this is the version of the view that's most popular, a social arrangement is only legitimate if no one could reasonably object to it.

This basic idea was very important in European thought through the 17th and 18th centuries, and hence to the Atlantic revolutionary movements over the late 18th and early 19th centuries.

Elkin and Pettigrew suggest that when we're ascribing views to a group, we should follow the same kind of contractualist logic.

It's only fair to ascribe an attitude to the group, in a way that matters for responsibility, if no one in the group could reasonably object to the ascription.

They suggest that accuracy consensus is plausibly a requirement for this kind of non-objection principle.

Is that right? Not sure. But I like the idea of thinking about the connections to political principles, and to the idea that we should make ascriptions that command a degree of universal assent.

We've mentioned a couple of times that it would be great if pooling methods could somehow merge everyone's private evidence.

When there are people who are better placed than us because they know more than we do, and we don't know what they know, but do know what credences they have, it would be great to have a pooling formula that took in all their evidence in the right way.

You can *sort of* do this with geometric pooling, but there's a massive catch.

If you know the full credal function of each of the experts, and you use geometric pooling with a maximally fine-grained partition, you get just what we wanted.

The result is as if you'd learned everything the experts know.

Unfortunately, to do this you need to know more than the expert's credence on the question at issue (e.g., polio eradication).

You also need to know their credence in various evidence propositions. And by looking at what they have credence 1 in, you can tell what their evidence is.

The geometric pooling doesn't actually help much.

Evidentialism

There is a very big picture question here about what the nature of testimonial knowledge is. Here are three very big picture views about how testimonial knowledge works.

1. **Sharing**: The speaker knows something, and shares it with the hearer.
2. **Indicator**: The speaker is a good measuring device, like a thermometer.
3. **Evidence Transfer**. The speaker saying something gives the hearer evidence that the speaker has evidence for what they are saying. This is usually enough reason for the hearer to believe it too.

Most philosophers like either option 1 or option 2. I kind of like option 3, and Elkin and Pettigrew here seem to like it too, but it's worth nothing this is a *very* unorthodox view in philosophy, compared to 1 or 2.

Summing Up

Trade-offs Are Inevitable

No single pooling strategy satisfies all desiderata:

- Linear pooling: good for accuracy consensus and diversity, bad for External Bayesianity and deference compatibility.
- Geometric pooling: good for External Bayesianity and maybe evidence pooling, bad for partition sensitivity and attributibility.
- Context determines which trade-offs are acceptable.

The expertise problem: How should we weight different individuals?

- Equal weights? (Democratic but potentially inaccurate)
- Expertise-based weights? (Accurate but who decides expertise?)
- This connects to social epistemology and democratic theory.

Philosophical Implications (cont.)

Peer disagreement revisited: These accuracy arguments might inform *individual* responses to peer disagreement.

If accuracy matters, and if linear pooling promotes accuracy, perhaps individuals should adopt linear pools of their own credence and their peers' credences?

But: Deference Compatibility problem suggests caution here.

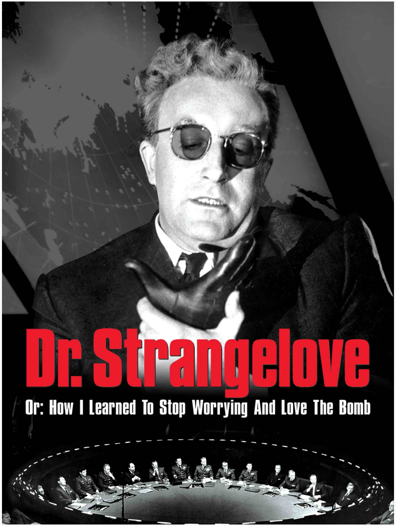

Figure 1: Movie poster for Dr. Strangelove

We're going to go back in time, to look at two famous contributions by the economist/political theorist/generally interesting person Thomas Schelling.